

HaloDB will be a place where a group of researchers could share the data related to halophil biologic samples. To do that they will have to access to a web application, the data they provide should be shared between members of a group of researchers or it can be made public, accessible to everybody.

This document contains the requirements established for the described web application.

Requirements for the general aspects of the web application

This section covers the aspects related with the users, the permissions they have to access data and the general aspects of the application. There is a specific section dealing with the requirements related to the data that HaloDB has to contain.

User Requirements

User requirements focus on what the end users expect to achieve with the web application. They describe the functionalities and features from the user's perspective.

1. User Registration and Authentication:

- **User Registration:** Users should be able to create an account by providing essential information such as username, email address, and password and also the research group they belong.
- **Email Verification:** Require users to verify their email addresses during registration to enhance security and reduce fake accounts.

2. Limited Access for Unregistered Users:

- **Access for Unregistered Users:** Unregistered users will be able to access only to public data.

3. User Profiles and Personalization:

- **User Profiles:** Registered users will be able to create and edit their profiles with personal information, avatars, and preferences.
- **Personalized Content:** Use user data to personalize content recommendations, notifications, and user experiences.

4. Group Creation and Management:

- **Group Creation:** Registered Users will be able to create groups with customizable settings, such as group name, description, and privacy settings.
- **Group Membership:** Registered Users will be able to invite others to join their groups or request to join existing groups.
- **Group Permissions:** A Registered User will be able to define granular permissions for group members within the group (who can post, edit, or delete content).

5. Notifications and Communication:

- **Notifications:** The web application will implement a notification system to keep users informed about their group activities, messages, and updates.

- **Messaging:** Registered Users will be able to communicate within groups through messaging features (chat, discussion boards).

6. Content Creation and Sharing:

- **Content Types:** The information that will be dealt with in the web application is described in the corresponding section.
- **Sharing:** Registered users will be able to share content within their groups or with specific individuals.
- **Public access:** Registered Users will be able to publish the data they have with Owner Permission.
- **Restricted access:** The users will have some permissions granted, and they will be able to interact with the data according to the permissions they have.
- Registered Users should be able to upload raw data files.
- Registered Users should be able to fulfil the complementary information describing the data.
- Permissions should control who can upload, edit, and delete data and documents. **NOTA: Meditar sobre l'eliminació de les dades públiques.**
- The ownership of the data should be transferred from one Registered User to another.
- Registered Users belonging to a group should be able to collaborate on data and associated documents (comments, annotations, or discussion threads).
- **Data validation:** The files uploaded should be validated.

7. Data Download:

- Registered Users should be able to download raw data files and associated documents, if they have the corresponding permission.
- Download options should include different file formats if applicable.

8. Data import and export:

- Registered Users will be able to import data from (ENA/EMBL/NCBI).
- Registered Users will be able to export data to (ENA/EMBL/NCBI).

9. User Feedback and Support:

- **Feedback Mechanism:** *Include a way for users to provide feedback, report issues, or request assistance.*
- **User Support:** The application will offer customer support channels to assist users with problems or inquiries (*email, tutorials,*).
- Users should receive notifications about updates, comments, or changes to data and documents.

10. Search and Retrieval:

- Users should be able to search for data and documents based on various criteria (e.g., keywords, date, sample information,).
- Search results should be relevant and organized.
- Search results will be restricted by the permissions the user has on the resulting information.

11. Version Control:

- Support for versioning data and documents to track changes and revisions.

Functional Requirements

Functional requirements describe specific functionalities and features that the web application should have. They focus on what the system should do.

1. User Roles and Permissions:

- **Define User Roles:** There will be different user roles (e.g., admin, registered user, visitor) with varying levels of access and permissions
- **Access Control:** *Specify which features and content are accessible to each user role.*

2. Group Permissions:

- **Define Granular Permissions:** A user belonging to a group will have a defined access permission (post data, edit data, read-only data and delete data)
- **Access Control:** A user can only access group data according to the permissions they have.

3. User Authentication:

- **Password Security:** The web application will implement strong password requirements and will provide security measures like password hashing and encryption.

4. File Upload and Storage:

- Allow users to upload various types of files (e.g., fasta, CSV, PDF, images, ...).
- Store files securely and efficiently, possibly in a database.

5. Data and Document Creation:

- Provide forms or templates for users to create information documents.
- Support document metadata (e.g., title, author, date) and structured data entry.

6. Search and Retrieval:

- Implement a robust search engine or database indexing system for efficient data retrieval.

7. Version Control:

- Track changes to data and documents, allowing users to view and revert to previous versions.

8. Collaboration Features:

- Implement features for user collaboration, such as commenting, annotations, or discussion threads.
- Group management with permissions for sharing and editing.

9. Data Download:

- Provide secure download options for raw data files and documents.
- Ensure file integrity during downloads.

10. Data Security:

- There will be access controls to protect the access to non-public information.
- There will be data backup and recovery mechanisms.

Non-Functional Requirements

Non-functional requirements describe qualities or characteristics that the web application should possess. They are often related to performance, security, and usability.

1. Performance:

- Ensure fast and responsive user interactions, especially when handling large files and datasets.

2. Security and Privacy:

- **Data Security:** Ensure the security of user data through encryption, secure connections (HTTPS), and regular security audits.
- **Privacy Controls:** Provide users with options to control their privacy settings, including who can see their profile and content.

3. Scalability:

- Design the application to scale as the volume of data and users grows.

4. Data Integrity:

- Ensure data integrity through checksums or validation mechanisms.

5. Backup and Disaster Recovery:

- Implement regular data backups and a disaster recovery plan to prevent data loss.

6. Accessibility:

- Make the application accessible to users with disabilities in compliance with accessibility standards (e.g., WCAG).

7. Analytics and Reporting:

- **Tracking User Engagement:** Implement analytics to track user engagement, group activity, and other relevant metrics for ongoing improvements.

- **Compliance:** Ensure compliance with data protection regulations (e.g., GDPR) and industry standards for user data privacy and security.

8. User Training and Support:

- Provide user training materials and support resources to assist users in using the application effectively.

Data Requirements

This section covers the specific data requirements

Source of data and classification

A researcher collects an environmental sample. There can be several types of data from the sample:

- A. Metagenomic data
 - a. Reads
 - b. Binning
 - c. MAGs, a file for each MAG
- B. Metaviromic data
 - a. Reads
 - b. Binning
 - c. MAGs, a file for each MAG
- C. Metatranscriptomic data
 - a. Reads
 - b. Binning
 - c. MAGs, a file for each MAG
- D. Metaproteomic data
 - a. Reads
 - b. Binning
 - c. MAGs, a file for each MAG
- E. Cultures
 - a. A file for each culture
- F. Virus isolation
 - a. A file for each culture?

Data representation

In general, the data must be in a format compatible with ENA standards. There must be a procedure capable to import and export data from the EMBL, NCBI, JGI repositories.

Incorporar documentació composta de reunions d'HaloDB

Tools

There are some functionalities that will be available, at least they will be:

1. Visualization
2. Statistical treatment of the data. Data analytics
3. Querying (BLAST)

4. Comparison.
5. MetaDAG
6. Meteor (+ VPF-Class, Prodigal)
7. Adding pipelines (from raw data to annotations)
8. In general: data handling and building derived products

User Requirements:

1. User Registration and Authentication:

- Only Registered users can upload data (See user requirements).
- Visitors can view the public data.

2. Data Upload and Management:

- The users will have some permissions granted, and they will be able to interact with the data according to the permissions they have.
- Registered Users should be able to upload raw data files.
- Registered Users should be able to fulfil the complementary information describing the data.
- Permissions should control who can upload, edit, and delete data and documents.
- The files uploaded should be validated.

3. Search and Retrieval:

- Users should be able to search for data and documents based on various criteria (e.g., keywords, date, sample information,).
- Search results should be relevant and organized.
- Search results will be restricted by the permissions the user have on the resulting information.

4. Version Control:

- Support for versioning data and documents to track changes and revisions.

5. User Collaboration:

- Registered Users belonging to a group should be able to collaborate on data and associated documents (comments, annotations, or discussion threads).

6. Data Download:

- Registered Users should be able to download raw data files and associated documents, if they have the corresponding permission.
- Download options should include different file formats if applicable.

7. Data import and export:

- Registered Users will be able to import data from (ENA/EMBL/NCBI).
- Registered Users will be able to export data to (ENA/EMBL/NCBI).

8. Notification and Alerts:

- Users should receive notifications about updates, comments, or changes to data and documents.

9. Data Visualization:

- Tools for visualizing data???

10. Data Security:

- There will be access controls to protect the access to non-public information.
- There will be data backup and recovery mechanisms.

Functional Requirements:

1. File Upload and Storage:

- Allow users to upload various types of files (e.g., fasta, CSV, PDF, images, ...).
- Store files securely and efficiently, possibly in a database.

2. Data and Document Creation:

- Provide forms or templates for users to create information documents.
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- Ensure file integrity during downloads.

Non-Functional Requirements:

1. Performance:

- Ensure fast and responsive user interactions, especially when handling large files and datasets.

2. Scalability:

- Design the application to scale as the volume of data and users grows.

3. Security:

- Implement robust security measures to protect data, including encryption, authentication, and authorization.
- Regular security audits and updates.

4. Data Integrity:

- Ensure data integrity through checksums or validation mechanisms.

5. Accessibility:

- Make the application accessible to users with disabilities in compliance with accessibility standards (e.g., WCAG).

6. Compliance:

- Ensure compliance with relevant data protection regulations (e.g. GDPR) and industry standards.

7. User Training and Support:

- Provide user training materials and support resources to assist users in using the application effectively.